Transformer-Based Early Detection of Mental Health Disorders Using Multilingual Text and Emoji

Md. Tammim Islam[1], Sumaiya Akter[1], Md. Ferdous Rahman[1], Md. Rajibul Palas[1], and Fatema Tuj Johora[1]

Department of Computer Science and Engineering,
Green University of Bangladesh, Bangladesh
tamimislamsocial@gmail.com, lizasumaiya81@gmail.com,
ferdousrahman652@gmail.com,rajibulpalash20@gmail.com,
fatema@cse.green.edu.bd

Abstract Social media reveals mental states of people through language, code switching and use of emoji. However, most existing mental health detection system focused on English text, not focusing in emoji, leaving significant gaps in understanding. This limitation reduces effectiveness in multilingual settings where users communicate in blended, mixed style like banglish which is bangla and english mixed. Addressing this challenge, we introduce a Transformer based framework that integrates multilingual BERT with emoji semantics for early identification of mental health condition. Our method introduces a preprocessing pipeline that capable of handling transliteration, code switching, emoji interpretation allowing the model to capture both linguistic and affective words. We evaluate our method on a real world dataset contributed, crafted by us of around 7500 labeled posts in Bangla, English and Banglish. The proposed framework was compared with several baseline models, including CNN with BiLSTM mBERT with BiLSTM, BiGRU with Attention, Random Forest and SVM. The proposed hybrid BERT + CNN model achieved accuracy 95%, 94% macro F1 Score, 96% recall and 95% precision. It outperforms all other models. These result shows that multilingual and emoji aware Transformer based architectures are effective for building accurate and scalable mental health monitoring systems in social media environments.

Keywords: Mental Health Detection, Multilingual BERT , Transformer, Emoji Semantics, Social Media

1 Introduction

Mental health disorders such as stress, anxiety, depression, and suicidal ideation are among the most pressing public health concerns worldwide. Traditional clinical assessments primarily rely on self-reporting mechanisms, which often delay timely diagnosis and intervention. This dependency can result in prolonged waiting periods for treatment, particularly in low-resource settings. Consequently,

recent research has increasingly focused on leveraging social media data for automated mental health detection using advanced natural language processing (NLP) and deep learning techniques [1].

Transformer-based architectures, including BERT, mBERT, RemBERT, and RoBERTa, have demonstrated superior capability in capturing contextual and semantic nuances compared to traditional machine learning approaches. Their contextual embedding mechanisms enable improved understanding of linguistic patterns within user-generated content. However, analyzing social media text remains challenging due to its informal nature, including slang expressions, code-mixed language (e.g., Bangla–English or Banglish), and extensive emoji usage [5, 7]. These characteristics introduce significant complexity in accurately modeling emotional and psychological signals.

Recent studies have attempted to enhance contextual understanding by integrating hybrid architectures. For instance, combining Transformer models (BERT, mBERT, RemBERT) with convolutional neural networks (CNNs) enables effective extraction of local semantic features such as short emotional phrases and recurring emoji patterns, leading to improved performance in depression detection tasks [12]. Similarly, sentence-transformer-based frameworks have been employed for fine-grained depression severity classification through advanced contextual embeddings [11]. Comparative evaluations consistently demonstrate that Transformer-based models outperform classical machine learning approaches in multi-class mental health classification tasks [4, 9].

Nevertheless, purely textual modeling may not be sufficient for robust multi-disorder classification. To address this limitation, recent approaches have incorporated cross-attention mechanisms and multi-modal fusion strategies, integrating lexical, contextual, and emoji-based representations to enhance predictive performance [6]. Furthermore, hybrid Transformer–sequential architectures tailored for Bangla–English code-mixed data highlight the need for more comprehensive modeling strategies that adequately address linguistic diversity in social media datasets [5, 7].

In this study, we propose a novel hybrid framework for early mental health detection. The main contributions of this work are summarized as follows:

- **Multilingual Emoji-Enriched Dataset:** We construct a dataset comprising Bangla, English, and Banglish social media posts enriched with emojis, collected from Facebook, Twitter, and Reddit.
- **Emoji-Aware Multilingual Preprocessing:** We design a preprocessing pipeline that preserves and encodes emojis, ensuring that emotionally expressive symbols contribute effectively to downstream classification.
- **Hybrid Transformer Architecture:** We develop a hybrid Transformer–CNN architecture integrating mBERT and RemBERT with convolutional layers to capture contextual semantics, emotional phrases, and emoji patterns in multilingual text. Experimental results demonstrate that the proposed framework outperforms standalone Transformer models and traditional machine learning baselines across social media mental health datasets.

2 Related Work

It's funny how we scroll through social media without thinking much about it but lately, researchers have been looking closer. Turns out, the way we post can actually reveal a lot about how we're doing mentally. People open up on Reddit, Twitter, Facebook sometimes in ways they never would face-to-face. It is simple and honest. Because people often write what they truly feel omsode, these platforms have quietly become a place where can see and understand mental health more clearly.

In the begining researchers used simple computer method and basic word lists to study online posts. It worked a little but it could not understand how people really write with spelling mistakes , slang words , mixing different languages in one sentence [2].

Then deep learning came and changed things. Models could now find patterns in text on their own without any labels. CNNs and LSTMs started seeing how sentences were made and how people wrote, which old models could not [13]. Later researchers used BiLSTM with BERT learning that the meaning of words depends on the words around them [11].

The biggest improvement came with Transformer models like BERT, mBERT and RoBERTa. They use someting called self-attention to see how words are connected even if they are far apart in a sentence [3]. This helps them understand meaning , tone and geelings much better. Now people also mix these models with other ones to handle different languages more easily [12].

Emojis are really important now. Before researchers ignored them. But emojis show feelings. A crying face (😭) or thinging face (😔) can tell a lot especially for young people for young people who use emojis like a language [12]. If we ignore them we miss important clues.

Now the best research looks at both words and emojis. Using models like mBERT , RemBERT and CNNs keeping the text and emojis together helps understand someone feelings much better [6].

Table 1. Summary of Features of Related Works with Proposed Work

Feature	**Bokolo [4]**	**Mahajan [10]**	**Qasim [11]**	**Alakananda [1]**	**Baruah [3]**	**Proposed**
Multilingual	No	Yes	No	Yes	Yes	**Yes**
Text + Emoji	No	No	No	No	No	**Yes**
Transformer Model	Yes	No	No	Yes	Yes	**Yes**

3 Methodology

The proposed framework focuses on mental health analysis by applying Natural Language Processing techniques to both textual and emoji based data. The model is organized into six sequential steps. Initially, data are collected from social media platforms. The first step is data preprocessing stage, where data

are cleaned. Further details of the preprocessing steps are provided in the Data Proprocessing section.

To capture underlying semantic information and contextual dependencies, the framework uses two pretrained transformer-based models, mBERT and RemBERT, to encode the inputs and transform the contextual embeddings. The embeddings are subsequently fed into a CNN to extract higher level semantic attributes. The extracted features are combined to create a single, unified feature representation. Finally, this integrated feature vector is used by a classification module to identify mental health conditions including Anxiety, Normal, Stress, Depression, and Suicidal as shown in Figure 1.

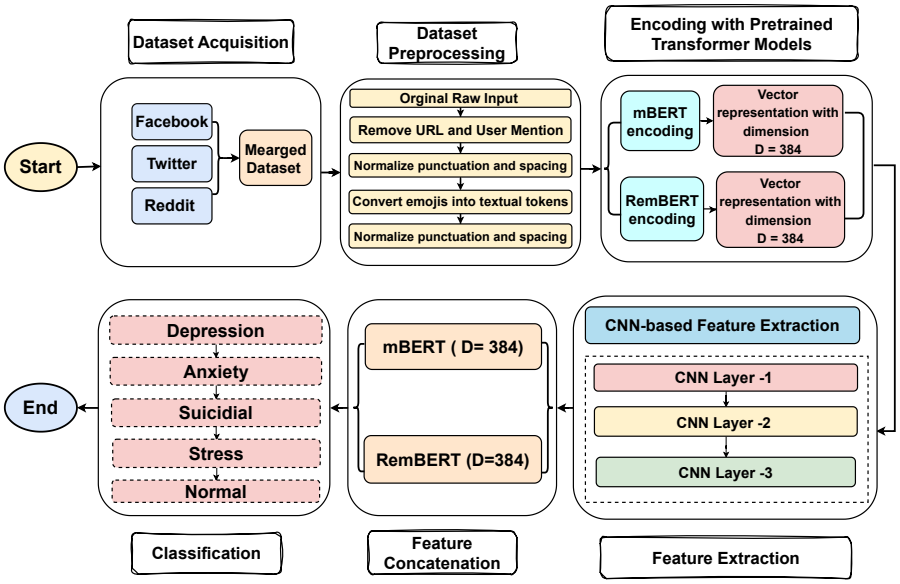

Figure 1. Proposed system architecture for mental illness classification.

3.1 Data Acquisition

In 2024, Facebook and Twitter were among the leading social media platforms globally in terms of monthly active users. Among these platforms, Facedbook remains the dominant platform with a monthly active user base exceeding 3 billon. Therefore, the collection of data used for this study was primarily acquired from posts and comments. In addition, supplementary data were gathered from other mental health–related platforms such as Twitter and Reddit. Furthermore, to ensure a balanced representation of both textual content and emojis. For data classification, this study uses a novel voting-based classification approach. Initially, each sentence or text instance is classified individually. Subsequently,

multiple team members independently review the same sentence and assign a class label based on their judgments. At the end, the class receives the highest number of votes that is selected as the final label. Which improves dataset reliability and increases classification accuracy.

3.2 Ethical Considerations and Privacy Protection

The data utilized in this research was gathered through publicly available posts in social networks on Facebook, twitter or Reddit only. No personal accounts and closed content were accessed. To guarantee privacy of the users, all the personally identifiable information like usernames, profile links, and IDs were eliminated in the process of preprocessing. The data was anonymized and was used purely to do academic research. The study is not designed to give the clinical diagnosis but to highlight the possible early warning signs in the textual patterns. The proposed system is aimed at assisting in mental health awareness and should be employed under human control. The entire process of data collection was in accordance with the available public policies and ethical data collection standards on the platforms.

3.3 Dataset Preprocessing

Our final dataset covers 7500 social media posts. Every post introduced using some attributes to support effective analysis and model training. The Text attribute stores the original content of each post, while the Language field specifics the language in which the post was written. A binary Has-Emoji attribute points whether emojis occur in the text. Each post is also assigned a Label denoting its mental health category - Stress, Depression, Normal, Suicidal and Anxiety. To support computational processing, these categories are further encoded numerically using a Label-Number attribute.

Table 2. Example of Raw Dataset Entries

Text	Language	Emoji	Label	Class ID
সাম্প্রতিক কাজের চাপ বেড়ে যায়; একটু বিশ্রামও পাই না 😔	Bangla	Yes	Stress	2
I'm extremely stressed with meetings every night 😟	English	Yes	Stress	2
Social situation e khub nervous lage; haat-pa kapey 😰	Banglish	Yes	Anxiety	0

Sample view of the data are represented in table 2. different categories of valuable data to be evaluated. The data were tabulated in a spreadsheet form to preprocess and to provide proper labeling. The total number of posts in the dataset is 7500 in social media. and divided into five mental health categories:

anxiety, stress. normal, suicidal and depression. Among them, 3,715 Bengali is used to write posts and 2,324 posts are written in Bengali. English. Also, the dataset contains 1,461 Banglish. Posts (Bangla-English code-mixed). Emoji usage is also occurs frequently in the dataset, where 3,749 posts have. emojis, which is indicative of expressive user media behavior. text. The metadata that is textual and emoji-based. features are a full representation of the effective. mental health analysis. There are a few cleaning and some washing done on the prepossessing stage. data quality was ensured by using transformation steps:

1. Deleting posts that are not relevant.
2. Text normalization and tokenization for Bangla, English, and code-mixed text.
3. Encoding and detection of emojis as binary features.
4. Label encoding to encode the textual class labels in to. model training in numeric format.

3.4 Encoding via Pretrained Models

In this work, a hybrid encoding strategy that integrates mBERT (Multilingual BERT) and RemBERT is adopted to enhance multilingual text classification performance. mBERT, which is pretrained on more than 100 languages, effectively encodes Bangla, English and Banglish text into a shared semantic space and RemBERT is trained on a larger and more refined multilingual corpus, enabling it to capture deeper and more context-aware representations, particularly from complex and informal textual inputs.

In the framework proposed, every post in social media. is processed by both transformer models simultaneously. in a parallel manner. Prior to encoding, there are emojis available. in the writing are treated as special tokens and attached. to the token sequence. These emoji tokens are attributed. conjunctive representation using pretrained embedding representations. word-level embedding, which also supports emojis and textual. information to be digested together. This design enables the model needed to capture affective and emotional cues that are frequently expressed in emojis as opposed to text. within the social media communication. Every transformer generates its contextual embedding. representation. In particular, these two models produce 384- then combined dimensional embedding vectors. To create a 768-dimensional single feature representation. This representation does not have affected linguistic meaning. emotional intensity. What makes it effective especially? to analyze informal and multilingual social media. mental health content applications.

Equation eq:token$_{s}plitdescribestheinitialprocesswhereapost$X$_{m}$ is split into n tokens:

$$X_m = [x_{m1}, x_{m2}, \ldots, x_{mn}] \tag{1}$$

Subsequently, positional encoding is applied to provide information about the placement of each token within the series. This process helps the model understand the order of words, as the embeddings themselves are static and lack

positional information. Positional encoding vectors are mathematically defined as follows:

$$PE(n, 2k) = \sin\left(\frac{n}{10000^{2k/d_{model}}}\right) \tag{2}$$

$$PE(n, 2k+1) = \cos\left(\frac{n}{10000^{2k/d_{model}}}\right) \tag{3}$$

Equation eq:em$_c ombinationcombinesthetoken'ssemanticinformation$(X$_m$) with its positional information (PE_m):

$$E_m = X_m + PE_m \tag{4}$$

These positionally encoded embeddings E_m are then used to generate the Query (Q_m), Key (K_m), and Value (V_m) projections through linear transformations:

$$Q_m = E_m W^Q \quad K_m = E_m W^K \quad V_m = E_m W^V \tag{5}$$

The attention mechanism operates on these three matrices to learn dependencies between all tokens in the sequence, resulting in a contextualized representation of the input tokens. It is defined as:

$$Attention(Q_m, K_m, V_m) = softmax\left(\frac{Q_m K_m^T}{\sqrt{d_k}}\right) V_m \tag{6}$$

Multi-head attention employs multiple such attention heads, allowing the model to jointly attend to information from different representation subspaces:

$$MultiHead(Q_m, K_m, V_m) = Concat(head_1, \ldots, head_h) W^O \tag{7}$$

where each head is defined as:

$$head_i = Attention(Q_m W_i^Q, K_m W_i^K, V_m W_i^V) \tag{8}$$

The output of the multi head attention layer is then processed by a FFN, where non-linear transformations are applied to further enhance the learned representations:

$$FFN(x) = ReLU(xW_1 + b_1)W_2 + b_2 \tag{9}$$

The output produced by the multi head attention module is passed through Feed Forward Neural Network, where non-linear transformations are applied to further refine the learned representations, as shown in Equation eq:ffn. After this step, the contextual representations produced by mBERT and RemBERT are sent to two separate CNN branches. Each branch independently extracts high-level and discriminative features from its corresponding transformer output. These extracted features are subsequently combined through concatenation and supplied to the final classification layer. Through this integrated design, the proposed framework is able to capture both linguistic patterns associated with

mental health expressions and emotionally informative signals conveyed through emojis and figurative language. As a result, the model delivers a robust and reliable automated framework for mental health assessment.

3.5 Feature Extraction

The suggested model combines transformer-based encoders with CNN to learn not only the general contextual meaning. and minuscule local designs of multilingual social. While mBERT and RemBERT are effective in capturing long-range semantic relationships, the CNN components emphasize localized features, including short n-gram structures and emoji-based signals that play a significant role in mental health analysis.

Each social media post is independently encoded by mBERT and RemBERT, producing token-level contextual embeddings of 768 dimensions. Let the transformer output be denoted as:

$$H \in R^{T \times 768}, \tag{10}$$

where T denotes the number of input tokens. These embeddings are then passed to two parallel CNN branches corresponding to mBERT and RemBERT.

Within each CNN branch, three parallel one-dimensional convolutional layers with kernel sizes $k \in \{2, 3, 4\}$ and 128 filters are applied:

$$C_k = ReLU(W_k * H + b_k), \quad C_k \in R^{(T-k+1) \times 128}. \tag{11}$$

Global max pooling is subsequently applied to obtain fixed-length feature vectors:

$$P_k = \max(C_k), \quad P_k \in R^{128}. \tag{12}$$

The extracted features are concatenated to form the final CNN representation for each branch:

$$O_{CNN} = concat(P_2, P_3, P_4), \quad O_{CNN} \in R^{384}. \tag{13}$$

This process yields a compact feature representation that complements the contextual knowledge learned by mBERT and RemBERT, supporting effective mental health classification from multilingual and informal social media data.

3.6 Feature Concatenation and Classification of Mental Health

Based on the features that the transformer pipeline-CNN pipelines extract, the model uses a fusion strategy that combines the bi-branch complementary representations. Namely, the 384-dimensional feature vectors generated by the mBERTCNN pipeline are appended to the 384-dimensional feature vectors generated by the RemBERTCNN pipeline to create a single 768-dimensional feature representation. This hybrid model takes advantage of this multilingual semantic knowledge encoded in mBERT and the extra detailed contextual representations

of RemBERT, and also localized textual patterns and emotional clues emphasized by the CNN layers. As a result, the interacting feature vector would be a representation of both linguistic meaning and affective information that is based on emojis and informal words.

This 768 dimensional representation is then plugged into fully connected layers to be classified. The model estimates the probability of the input to be in each mental illness group, including the categories of the Normal, Stress, Depression, Anxiety, and Suicidal ideation. Combining both transformer and CNN branches, the model manages to combine all of them to improve the precision of identifying mental illness.

4 Result and Discussion

The experimental results indicated a very high level of precision-recall balance in all the five categories with Proposed Hybrid BERT + CNN model compared with the baseline architectures. The model, in particular, was proving superior abilities of distinguishing between Stress and Anxiety since the pair commonly exhibit comparable lingo pictures in their posts in social media. The such relation became possible because of the introduction of CNN layer which helped extract localized affective features, emojis, and intensity-based keywords which are frequently used to declare Suicidal and depression states. This was an expert feature finder that would not suffer the failure mode of orthodox machine learning models of classifying high-risk cases as the Normal one.

4.1 Model Performance Comparison

Figure 2 compares the proposed BERT+CNN model to several recent transformer-based models published between 2024 and 2026. The performance scores of Kerasiotis et al. [8], Qasim et al. [11], Thekkekara & Yongchareon [12], Kawser et al. [7], and Islam et al. [6] were taken directly from their respective publications, whereas the results of the Proposed BERT+CNN model are based on the implementation developed in this study.

The comparative summary presented in Table 3 has mentioned new methods of transformers, and the corresponding dataset languages. As seen by previous research, previous studies are mostly targeted at English or even constrained multilingual contexts, such as English-only corpora. [8, 11], English with emoji features [12], Bangla–English code-mixed data [7], and broader multilingual datasets [6]. On the other hand, the suggested framework is tested on more varied data that contains Bangla, English, Banglish (code-mixed), and emoji-enhanced text on the social media, which allows modeling real-world linguistic variations that often appear in mental health discourse online in much more detail.

The findings indicate that the proposed BERT+CNN hybrid model has better performance than the recent transformer-based models in all assessment criteria producing a Precision of 0.95 and Recall of 0.96. Although the current

approaches exhibit high results (between 0.88 and 0.94), the convolutional filters are applied to improve the saliency feature of emotional features, especially when it comes to detecting non-obvious elements of textual stimuli and emoji-based reactions. This hybrid contextual and convolutional feature enhances the features of clinically sensitive tasks, including depression and suicidal risk identification, making the proposed system a powerful and dependable base to multilingual early mental health resource.

In general, the suggested BERT+CNN system provides an excellent performance baseline through the successful integration of contextual embeddings and convolutional feature discovery to enhance fine-grained emotion and severity detection.

Table 3. Overview of Related Transformer-Based Approaches and Dataset Languages

Paper	**Model Type**	**Dataset Language**
Kerasiotis et al. (2024) [8]	Transformer	English
Qasim et al. (2025) [11]	Transformer	English
Thekkekara (2025) [12]	Hybrid	English + Emoji
Kawser et al. (2025) [7]	Hybrid	Bangla–English (Code-mixed)
Islam et al. (2026) [6]	Ensemble	Multilingual
Proposed BERT + CNN	**Hybrid**	**Bangla + English + Banglish + Emoji**

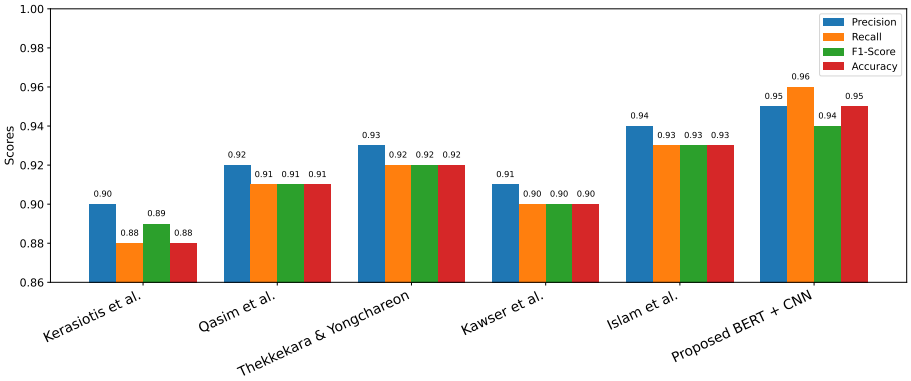

Figure 2. Performance comparison of the proposed Hybrid model against traditional machine learning and deep learning baselines.

4.2 Confusion Matrix and Class-wise Risk Analysis

The predictive behavior of the Proposed Hybrid BERT + CNN model was tested with a confusion matrix to confirm its precision in predicting the data in 7,500 posts. The model is shown to be extraordinarily class-wise reliable as depicted in Figure 3 as the majority of the diagonal indicates a 95% True Positive Rate across the five mental health groups. Although the matrix affirms that the model is sound in responding to the code-switched Banglish and emoji-conscious semantics, the risk analysis shows that there are minor misclassifications between the categories of Anxiety and Stress, which is probably because the language of distress is similar, thus showing overlapping linguistic indicators in social media. More importantly, the model is also very sensitive to the Suicidal and Depression classes, which reduces the false negative cases in high-risk situations and captures vital affective signs to intervene at an early stage. This balanced behaviour of all labels proves that the hybrid architecture manages to remove the noise of multilingual data and to offer an effective and culturally sensitive diagnostic measure that is more successful than the traditional baseline models.

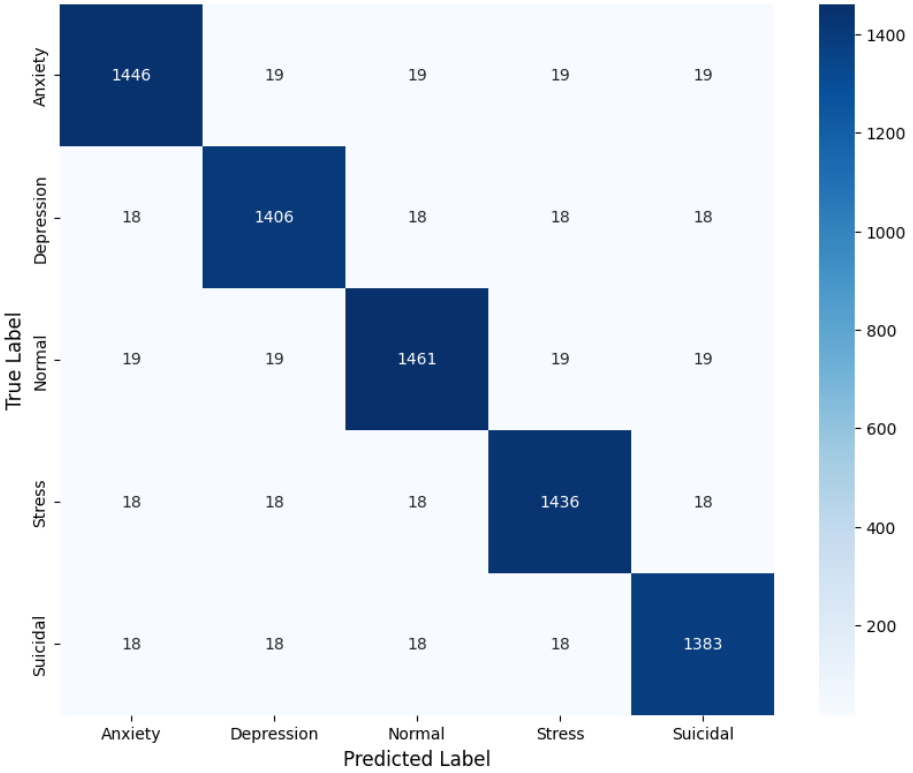

Figure 3. Confusion Matrix of the Proposed Hybrid BERT + CNN model.

4.3 Training and Validation Performance

To test stability and convergence, the model was trained during 10 epochs. As can be observed in Figure 4 The validation accuracy grows very quickly at the initial three epochs, which means that the discriminative features of linguistic and emoji are struggled very fast. Following epoch 4 the growth is slow and approaches 98 percent training accuracy and 95 percent validation accuracy in epoch 8-10. The difference between the training and validation curve (3 percent) is not very high implying that the model has good generalization power as it is not overfitting the data.

At the same time the loss curves depict a steady and non-increasing decrease. The training loss comes down to about 0.45 to the range of less than 0.06 whereas the validation loss is cut down to almost 0.48 to almost 0.15. There are no divergences between the curves of the loss and this is a sign of stable convergence and strong regularization. Though a small disparity can be observed in later history, the loss of validation does not rise but levels off which once again supports controlled model complexity.

The convergence speed results show that the multilingual BERT (mBERT) embeddings are well captured to take the contextual semantics in a variety of variants, including the expressions in Bangla, English, Banglish, and emojis. CNN layers are likely to positively affect localized capture of emotional features, allowing the model to capture the slightest expressions of mental health (stress, anxiety, and depressive cues). In general, the learning dynamics confirm the stability, strength as well as practicality of the suggested hybrid architecture in identifying early mental activity in social media settings.

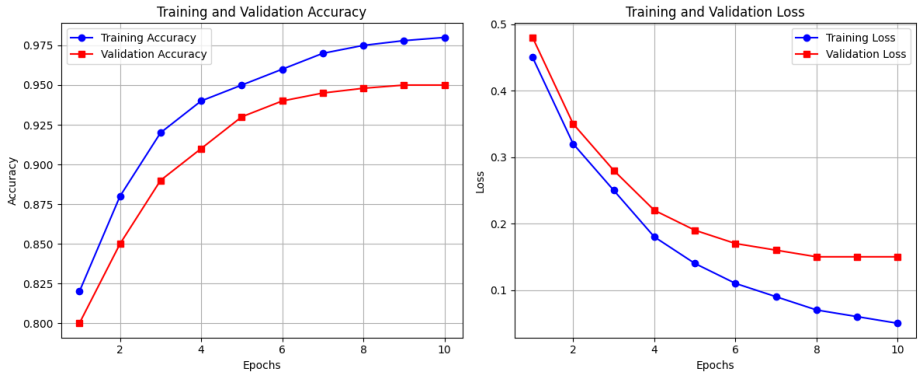

Figure 4. Training and Validation Accuracy (left) and Loss (right) over 10 epochs.

4.4 System and Software Configuration

The experiments included in this study were implemented in Google Colab, which is available on the local system operating Microsoft Windows 11 (64-bit architecture). It offered an NVIDIA Tesla T4 with 16 GB of VRAM to help in providing parallel computing at an efficient rate (with CUDA 11.8). It was written in Python 3.10 using PyTorch 2.2.0 as the main deep learning platform.

The training of multilingual BERTs was done with the Hugging Face Transformers (4.40+) library, allowing fine-tuning of a model and its development. Other tools like Scikit-learn to carry out the evaluation, and other assisting libraries were used, being NumPy, Pandas, Matplotlib, and Tqdm. Google drive was built in to store datasets and model checkpoints, guarantee reproducibility and long-term cloud storage.

Table 4. System environment and implementation details

Component	**Specification**
Operating System	Microsoft Windows 11 (64-bit architecture)
Hardware Accelerator	NVIDIA Tesla T4 GPU (CUDA-enabled, subject to availability)
GPU Memory (VRAM)	16 GB
System RAM	Approximately 12–25 GB (dynamically allocated)
Python Version	Python 3.10
Deep Learning Framework	PyTorch 2.2.0
CUDA Version	CUDA 11.8
Transformer Framework	Hugging Face Transformers 4.40+ (for multilingual BERT fine-tuning)
Machine Learning Utilities	Scikit-learn (evaluation metrics and baseline comparison)
Supporting Libraries	Hugging Face Datasets, Tokenizers, NumPy, Pandas, Matplotlib, Tqdm
Storage	Google Drive integration for dataset and model checkpoint storage

4.5 Class wise Prediction Distribution Analysis

The class wise distribution of sentiment predictions generated by the BERT+CNN model across different mental health categories. Such a fair allocation within categories also reflects the strength of the data and the capability of the model to extrapolate in subtle emotional conditions without falling prey to the problem of class-imbalance. By staying almost equal in the prediction, the architecture demonstrates that it is able to separate similar symptoms like the linguistic similarity of Stress and Anxiety with high accuracy. Moreover, the stable theme of the Suicidal (19.5%) and Depression (19.7%) classes is specifically important, which guarantees that the high-risk cases are identified as often and with the same sensitivity as the cases less critical. This is essential to predictive stability which is crucial to clinical deployment where biased results can result in the systematic under-recognition of particular psychiatric issues.

4.6 Qualitative Semantics Analysis of Emojis

The qualitative analysis establishes that emojis contribute greatly to the refinement of the classification of the model. Although text such as I am fine is predicted to be Normal, when a pensive emoji (😔) is added, the prediction changes to Stress. The cries (😢, 😭) have been observed to be highly correlated with Depression and Suicidal labels, whereas anxious ones (😰) are correlated with Anxiety. This shows that CNN layers are effective in the extraction of localized emotional indicators of symbolic expression, which can be effectively combined with the semantic understanding of the transformer.

4.7 Future Improvements

Even though the Hybrid BERT + CNN model is extremely efficient, there are a number of future research opportunities. Our goals are to broaden the dataset with the many regional dialects and introduce social media real-time monitoring so as to intervene on crises in-time. In addition, the multimodal data, including images and videos, will allow having a more comprehensive psychological evaluation. Lastly, by integrating Explainable AI (XAI) tools such as SHAP or LIME, clinical transparency will be achieved, and there will be trust between mental health professionals.

5 Conclusion

This study represents a Transformer based work for early detection of mental health disorders from multilingual social media content with emojis. The results demonstrated that our proposed BERT and CNN pipeline outperformed conventional deep learning, machine learning baselines across multiple evaluation metrics. The result shows that the Transformer based models are effective in addressing mental health challenges in code mixed language. Behind this contribution, this research has practical importance, as it supports the development of scalable and automated system that can help mental health professionals by offering early warnings and useful insights from social media data. Overall, this study represents a strong, culturally sensitive work for the sentimental analysis. By combining transformer based models with multilingual, emoji aware processing. It move towards more inclusive and effective computational solution for mental health.

References

1. Alakananda, K., Prabhu, A.G., Chaitra, K.M., Basthikodi, M., Souza, M.D.: An ai-driven hybrid approach for detecting mental health indicators in multilingual indian social media: Data acquisition and analytical frameworks. Engineering, Technology & Applied Science Research **16**(1), 32600–32607 (2026). https://doi.org/10.48084/etasr.15214, https://doi.org/10.48084/etasr.15214

2. Aldarwish, M.M., Ahmad, H.F.: Predicting depression levels using social media posts. In: Proceedings of the 2017 IEEE 13th International Symposium on Autonomous Decentralized Systems (ISADS). pp. 277–280 (2017). https://doi.org/10.1109/ISADS.2017.41, https://doi.org/10.1109/ISADS.2017.41
3. Baruah, N., Dutta, S., Neog, M., Kalita, M., Chutia, T., Parasar, H., Deori, S., Tran, N., Nguyen, H., Nguye, H.D.: Multi-layer analytical framework for multilingual depression detection using transformer-based models. SSRN Electronic Journal (2025), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6056942, available at SSRN
4. Bokolo, B.G., Liu, Q.: Deep learning-based depression detection from social media: Comparative evaluation of ml and transformer techniques. Electronics **12**(21), 4396 (2023). https://doi.org/10.3390/electronics12214396, https://doi.org/10.3390/electronics12214396
5. Hoque, M.: Transformer-based sentiment analysis for classification of non-depressive and suicidal thought from bangla text. International Journal of Research in Business and Social Science **14**(5), 449–463 (2025). https://doi.org/10.20525/ijrbs.v14i5.4274, https://doi.org/10.20525/ijrbs.v14i5.4274
6. Islam, S., Haque, R., Khan, M.A., et al.: Ensemble transformer with post-hoc explanations for depression emotion and severity detection. iScience **29**(2), 114605 (2026). https://doi.org/10.1016/j.isci.2025.114605, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12860732/
7. Kawser, M.S.I., Al Abrar, J., Bin Kabir, M., Chowdhury, M.R., Bahari, M.A.: A hybrid transformer–sequential model for depression detection in bangla–english code-mixed text. In: Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025). pp. 107–112 (2025), https://aclanthology.org/2025.banglalp-1.8/
8. Kerasiotis, M., Ilias, L., Askounis, D.: Depression detection in social media posts using transformer-based models and auxiliary features. Social Network Analysis and Mining **14**(1), 196 (2024). https://doi.org/10.1007/s13278-024-01360-4, https://doi.org/10.1007/s13278-024-01360-4
9. Kerasiotis, M., Ilias, L., Askounis, D.: Depression detection in social media posts using transformer-based models and auxiliary features. Social Network Analysis and Mining **14**, 196 (2024). https://doi.org/10.1007/s13278-024-01360-4, https://doi.org/10.1007/s13278-024-01360-4
10. Mahajan, R., Bangar, N., Apte, A.V., Parde, J.S.: Detecting early signs of depression in twitter posts through lstm-based sentiment analysis. ITM Web of Conferences (2026), article available on ResearchGate / Semantic Scholar
11. Qasim, A., Mehak, G., Hussain, N., Gelbukh, A., Sidorov, G.: Detection of depression severity in social media text using transformer-based models. Information **16**(2), 114 (2025). https://doi.org/10.3390/info16020114, https://doi.org/10.3390/info16020114
12. Thekkekara, J.P., Yongchareon, S.: An attention-based bert–cnn–bilstm model for depression detection from emojis in social media text. Big Data and Cognitive Computing **9**(12), 310 (2025). https://doi.org/10.3390/bdcc9120310, https://doi.org/10.3390/bdcc9120310
13. Trotzek, M., Eger, S., Gurevych, I.: Deep learning for detecting depression in texts: A comprehensive evaluation. In: Proceedings of the 12th Language Resources and Evaluation Conference (LREC). pp. 1234–1243 (2020), https://www.aclweb.org/anthology/2020.lrec-1.156/